Blackjack Advantage AI
Risk of Ruin and Bankroll Optimization Tool

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EGR 404 - Generative AI Tools

Problem

- Blackjack advantage play relies on complex math
- Risk of ruin and variance are difficult to calculate manually
- Most players rely on spreadsheets or guesswork
- No simple tool that combines simulation and AI explanations

Solution

- Built a Streamlit application in Python
- Uses Monte Carlo simulation to model blackjack sessions
- Simulates fluctuating true count instead of fixed edge
- Includes wonging (sitting out negative counts)
- Uses bet spread based on true count
- Integrates OpenAI API to explain results and give advice

Key Features

- User inputs bankroll, bet spread, hours, and simulations
- Calculates expected profit and risk of ruin
- Simulates realistic blackjack conditions
- Displays bankroll distribution graph
- Shows true count distribution
- Generates AI analysis of results

Demo

- Enter inputs (bankroll, bets, hours)
- Run simulation
- View results (profit, risk of ruin)
- Analyze graphs
- Read AI-generated explanation

Challenges

- Modeling realistic true count fluctuations
- Implementing wonging and bet spread logic
- Managing simulation performance for large inputs
- Integrating OpenAI API correctly

Conclusion

- Successfully built a working blackjack risk analysis tool
- Combined simulation with generative AI
- Provides both data and easy-to-understand explanations
- Demonstrates real-world application of AI tools